Supplementary material: Auditing Source-Label Coverage and Recovery–Retention Trade-offs in Cross-Dataset Land-Cover Segmentation

Yangxi Luo[1], **Liang Cheng**[2,*], **Jiawen Pan**[2]

[1]Fuzhou Polytechnic University, Fuzhou, China

[2]Fujian Agriculture and Forestry University, Fuzhou, China

S1. Scope and provenance

This supplement documents the primary experimental probe, archived native U-Net comparison, label mappings, resampling rules and diagnostic boundaries. It does not add a new algorithm claim. Numerical statements must be traced to the original execution JSON, per-run configurations and manifests. The manuscript revision retains a file-level input inventory and a fresh paper-to-evidence audit; this is a working draft, and the final reviewer-accessible archive and persistent repository record remain to be finalized.

S2. Frozen ontology and source-label relations

The frozen ontology contains built structure, transport surface, woody vegetation, herbaceous vegetation, cropland, bare surface, and surface water. The output name herbaceous vegetation is an internal codebook alias: FLAIR herbaceous vegetation and Semantic Drone grass are assigned to it, while OpenEarthMap rangeland is an approximate low-vegetation proxy that also includes bushes and shrubs. Cropland denotes agricultural fields or managed crop cover. This is an operational crosswalk for a bounded experiment, not an assertion that the native dataset definitions are universally interchangeable.

Table 1: Raw labels and their declared relation to the common ontology.

Dataset	Native raw label and definition	Declared relation	Boundary and role
LoveDA	building, road, forest, agriculture, barren, water	Exact operational support for six non-herbaceous leaves	Primary source
LoveDA	no raw herbaceous label	Absent	Coverage gap
FLAIR	ID 10, *herbaceous vegetation*[1]	Declared herbaceous relation	B1 supervision only; agricultural and plowed labels ignored
FLAIR	IDs other than 10	Ignored	Do not alter LoveDA-supported leaves
OEM	*rangeland*, including bushes/shrubs[2]	Approximate operational proxy	Target label is broader than pure herbaceous vegetation
OEM	*agriculture_land*[2]	Declared cropland relation	Managed crop cover remains cropland
OEM	*developed_space*	Coarser ancestor	Partial-label treatment only
Semantic Drone	*grass*[3]	Declared herbaceous relation	Non-cropland herbaceous second-target score

OEM denotes OpenEarthMap in this supplementary table.

The FLAIR mapping is anchored in the official original class table, which names value 10 *herbaceous vegetation*.[1] The operational relation uses the native OpenEarthMap *rangeland/agriculture_land* distinction[2] and the released Semantic Drone *grass* class.[3] This is a declared non-cropland vegetation proxy, not identical herbaceous coverage: OEM rangeland includes bushes and shrubs. Neither the three native annotation guidelines nor their geographic footprints are assumed equivalent. The retained OEM masks do not separate the shrub component, so grass-only performance cannot be reconstructed from these labels. Difficult boundary cases, including crop fields with herbaceous appearance, remain cropland when their target reference label is *agriculture_land*; the ordinary confusion matrix retains those errors. Exact operational-support decisions are protocol declarations tied to raw-mask support. They do not assert common imaging conditions, acquisition dates, or geographic coverage across the contributing datasets.

S3. Complete frozen prompts

Table 2: Prompt set from `configs/taxonomy_prompts_v0.yaml`.

Leaf	Prompt
`built_structure`	a satellite image of a building
`transport_surface`	a satellite image of a road
`woody_vegetation`	a satellite image of forest or trees
`herbaceous_vegetation`	a satellite image of grassland or rangeland
`cropland`	a satellite image of agricultural cropland
`bare_surface`	a satellite image of bare soil or barren ground
`surface_water`	a satellite image of water

The prompt set is identical for E0, B0, B0-half, B0-half-batch2, B1, all OpenEarthMap replays, and Semantic Drone inference. Separate prompt diagnostics did not select a headline result.

S4. Calibration and tiled inference

For normalized image-patch feature $\mathbf{z}_{hw}$, normalized text feature $\mathbf{t}_c$, and rank $r = 32$, the reported logit is

$$\ell_c(h, w) = \tau\, \mathbf{z}_{hw}^{\top} \mathbf{t}_c + \alpha \frac{\langle \mathbf{W}_I \mathbf{z}_{hw}, \mathbf{W}_T \mathbf{t}_c \rangle}{\sqrt{r}},$$

where $\mathbf{W}_I, \mathbf{W}_T \in \mathbb{R}^{32 \times D}$, τ, and α are trainable. Both CLIP encoders are frozen and features are detached before the adapter. Each raster is partitioned into non-overlapping 512×512 RGB tiles. Right and bottom edge tiles are zero-padded. Logits are bilinearly interpolated with the align-corners parameter set to false; after argmax, padded predictions are cropped to the original extent. There is no overlap, blending, test-time augmentation, or post-processing.

S5. Fixed-budget arms and duplicate-crop conditions

Table 3: Frozen arms relevant to the primary analysis.

Arm	Updates	Source examples per update	Role
B0	25,220	Two LoveDA crops	Primary baseline
B0-half	25,220	One LoveDA crop	Exposure-matched control
B0-half-batch2	25,220	One deterministic LoveDA crop duplicated	Formal integrity check
B1	25,220	One LoveDA crop plus one FLAIR raw-ID-10 crop	Auxiliary source-mixture arm

For B0-half-batch2, duplicate copies of the same deterministic crop have equal individual losses. With mean loss, a frozen backbone, no train-mode batch-dependent state, no sample-dependent random transformation at the adapter, and identical optimizer state, the mean gradient equals the

one-crop gradient at every update. The absolute mean-mIoU contrast is below 4×10^{-9}, numerically zero at the reported precision, and is therefore a constructive pipeline-integrity check. It is not a generic batch-size estimate.

S6. OEM estimator and uncertainty

For each seed and city draw, raster confusion matrices are summed within city and sampled city confusions are pooled. Per-leaf IoU and mIoU are calculated from the pooled matrix separately per seed. The reported point estimate is the arithmetic mean of three B1 seed metrics minus the arithmetic mean of three matched B0 seed metrics. The same operation is conducted on 2,000 shared city draws (NumPy seed 20260803); the 2.5th and 97.5th percentiles form the paired city-bootstrap interval. Pairing-control contrasts use the separately recorded NumPy bootstrap seed 20260806; the primary source-mixture and exposure contrasts use 20260803. These intervals condition on completed checkpoints. The reported seed range is separate information about checkpoint variation.

Supported-leaf mIoU is

$$\mathrm{mIoU}_{\mathrm{sup}} = \frac{1}{6} \sum_{c \in \mathcal{C} \setminus \{c^*\}} \mathrm{IoU}_c.$$

No prediction or reference pixel is removed from the confusion matrix. Predicting herbaceous for a cropland reference pixel remains a cropland false negative and a herbaceous false positive in the ordinary one-vs-rest definitions.

S7. Per-leaf OpenEarthMap contrasts

The following 95% intervals are unadjusted descriptive paired city-bootstrap intervals. Herbaceous/rangeland is the primary leaf; the remaining leaves expose the allocation pattern and are not fourteen independent confirmatory tests.

Table 4: B1 minus B0 IoU on the 74-city OpenEarthMap population.

Leaf	Estimate	Lower	Upper
Built structure	-0.01958	-0.04881	+0.00555
Transport surface	-0.1482	-0.1845	-0.1134
Woody vegetation	-0.1554	-0.1880	-0.1225
Herbaceous vegetation	+0.3118	+0.2821	+0.3399
Cropland	-0.2005	-0.2497	-0.1498
Bare surface	-0.05080	-0.1094	-0.00103
Surface water	-0.03249	-0.07099	+0.00676

The cropland decrease is the largest observed covered-leaf change. The data do not identify whether it follows from a changed herbaceous–crop decision boundary, source-image distribution, sampling, loss weighting, or another interaction. None of these alternatives is collapsed into a capacity explanation.

S8. B2 protocol and B2-aligned replay

The B2 direct-supervision control uses a 70%/15%/15% hash-defined city split, trains only on optimization cities, and evaluates on held-out test cities. The test population contains 16 cities and 595 rasters. The SHA-256 prefix for the unordered test-city-name set is `197a41e3`; the corresponding test-pair-identifier-set prefix is `94767908`. These are set fingerprints, not hashes of a manifest file; the full digests are retained in the archived city-split record. Its held-out rangeland IoUs are 0.3934, 0.3890, and 0.3776 for seeds 19, 37, and 73 (mean 0.3867; sample SD 0.00815); the corresponding mIoUs are 0.5207, 0.5291, and 0.5108 (mean 0.5202; sample SD 0.00915). Its paired city-bootstrap rangeland contrast against frozen E0 is +0.3763 (95% evaluation-unit bootstrap interval [+0.3430, +0.4030]). This direct-supervision result is a learnability control, not an external-transfer result. B2-aligned E0/B0/B1 replays do not train; they apply frozen external checkpoints to this manifest. The manuscript's E0/B0 result is checkpoint-specific and does not enter the primary 74-city estimate.

S9. Semantic Drone exposure sensitivity

The grass score treats every non-grass reference color as a negative and counts false-positive grass predictions on those pixels; it does not average separate scores for the other classes. The primary Semantic Drone analysis uses the frozen 400-raster manifest and three completed B0/B1 checkpoint pairs. Its B1 grass IoUs are 0.2282, 0.2305, and 0.2228 (mean 0.2272; sample SD 0.0040); B0 is zero for all three checkpoints. The archived aggregate TP/FP/FN values support these three points and their sample SD. Full per-raster records for checkpoints 37 and 73 were not recovered, so their archived multi-checkpoint interval is not reported or claimed as independently reconstructed. Four rasters (`000–003`) were viewed during pipeline smoke. The 396-raster analysis excludes these identifiers but is exploratory because the current record does not independently establish a pre-result exclusion timestamp; it remains the archived seed-19 sensitivity. The paired bootstrap resamples rasters rather than training runs or sites.

S10. Shuffle schedule audits

Table 5: Schedule-audit summary for shuffled auxiliary masks. The recorded configurations specify density-multiset preservation and no native pairs. Primary density values below are preserved run-configuration metadata; the complete primary schedules were not recovered for independent revalidation. The second-case shuffle is retired because its post-run per-pair density mismatches do not support a density-matched pairing interpretation. No numerical per-pair tolerance is documented in the archived configurations; the values are descriptive audit records.

Case	Seed	Mean $\lvert\Delta d\rvert$	Max $\lvert\Delta d\rvert$	Status
LoveDA–FLAIR	19	0.001258	0.2135	pairing contrast retained
LoveDA–FLAIR	37	0.001235	0.2514	pairing contrast retained
LoveDA–FLAIR	73	0.001260	0.1910	pairing contrast retained
Semantic Drone–LoveDA	19	0.068109	1.0000	retired
Semantic Drone–LoveDA	37	0.300807	1.0000	retired
Semantic Drone–LoveDA	73	0.222407	1.0000	retired

No numerical per-pair density tolerance is documented in the archived configurations. The primary contrast is a configuration-based sensitivity; same-region derangement, density-bin matching and

no-native-pair constraints have not been independently revalidated from complete schedule bytes. The second-case schedule did not provide an equivalently interpretable pairing control after its output audit. It is therefore excluded conservatively rather than used to make a pairing or efficacy claim.

S11. Supporting numerical diagnostics

Table 6: Three-checkpoint Semantic Drone second-target evaluation and B2 direct-supervision learnability diagnostic. The Semantic Drone point estimate averages three completed checkpoints; the three-checkpoint result is descriptive, with no independently reconstructed multi-checkpoint raster interval. The plus/minus value is the sample SD across three completed checkpoints, not a standard error. B2 intervals are paired city-bootstrap intervals over 16 held-out cities.

Benchmark	Arm / contrast	Estimate [95% interval where reported]	Units	Role
Semantic Drone	B1 − B0 (3 checkpoints)	+0.2272 ± 0.0040	400 paired rasters	second-target evaluation
Semantic Drone	E0 grass IoU	0.0224	400 rasters	checkpoint diagnostic
Semantic Drone	B0 grass IoU	0.0000	400 rasters	checkpoint diagnostic
B2	rangeland IoU (3 seeds)	0.3867	16 held-out cities	learnability control
B2 held-out	B0 rangeland IoU	0.0000	16 held-out cities	baseline replay
B2	E0 rangeland IoU	0.0104	16 held-out cities	baseline diagnostic
B2	B2 − E0 rangeland	+0.3763 [+0.3430, +0.4030]	16 held-out cities	learnability contrast

S12. Archived native U-Net records

The native U-B0, U-B0-half and U-B1 evaluations correspond to three matched seeds and the common 2,638-raster, 74-city Paris-excluded OpenEarthMap population. Per-raster confusion matrices sum to each archived city matrix and metric; identifier sets and reference-class marginals match the retained primary records. These are integer-record checks, not pixel-mask or checkpoint replay. The saved training configurations record an ImageNet-initialized ResNet-34 encoder, a GroupNorm U-Net decoder and fixed encoder BatchNorm running statistics. The nine archived final checkpoint files were subsequently recovered and independently SHA256-matched to their original evaluation configurations and unique regular archive payloads. Initial model states, historical executed-code identity and model inference replay remain unverified.

The source-audit content is recovered by its recorded SHA-256: the available August-3 ready record has the digest recorded for the August-17 path in all nine native configurations. This equality recovers content, not the original pathname, date or proof of consumption. All nine training histories reach the recorded 25,220 updates. The native cohort supports the within-family source contrasts in Table 7. Historical shuffled arms and their failed/adapted schedule gates remain archived

and are excluded from the present efficacy analysis. No shuffled outcome is used to select a native version.

Table 7: All native U-Net checkpoint metrics reconstructed from archived integer records. The supported metric averages the six LoveDA-supported leaves. Values are descriptive fixed-checkpoint metrics; city intervals in the main text do not estimate training-seed uncertainty.

Arm	Seed	Rangeland IoU	Supported mIoU	Overall mIoU
U-B0	19	0.0000	0.2500	0.2143
U-B0	37	0.0000	0.2299	0.1971
U-B0	73	0.0000	0.2188	0.1875
U-B0-half	19	0.0000	0.2257	0.1935
U-B0-half	37	0.0000	0.2439	0.2090
U-B0-half	73	0.0000	0.1925	0.1650
U-B1	19	0.2957	0.1181	0.1435
U-B1	37	0.2767	0.0787	0.1070
U-B1	73	0.2762	0.1107	0.1344

S13. Subsequent archival asset recovery

The nine native U-Net final checkpoint files were recovered from the retained cloud archive. An independent byte-level check matched every file to its original evaluation-configuration SHA256 and unique regular archive member. The nine configuration files and three inspected source files also match archived bytes. The recovered checkpoints are prepared in a separate local archive; this does not attest historical initialization or actual execution of those code bytes and does not constitute a new inference replay.

The retained Semantic Drone provider ZIP contains the semantic masks for the fixed 400 historical identifiers. A separate read-only mask-count check finds the same grass-positive integer count as historical TP plus FN for every identifier, with 1,918,375,679 grass-positive pixels in total. The check uses the released grass color $(0, 102, 0)$ and records member digests and image dimensions. Count agreement is not pixelwise historical-mask identity and does not recover the missing checkpoint-37/73 prediction records; the corresponding multi-checkpoint raster interval remains withdrawn. No model or optimizer was run in these recovery checks.

References

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2 J. Xia, N. Yokoya, B. Adriano, *et al.*, "OpenEarthMap: A benchmark dataset for global high-resolution land cover mapping," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 6243–6253 (2023). https://doi.org/10.1109/WACV56688.2023.00619.

3 Graz University of Technology, "Semantic drone dataset." Dataset. Accessed 2026-10-15; project duration 2018–2019; https://ivc.tugraz.at/research-project/semantic-drone-dataset/.